Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

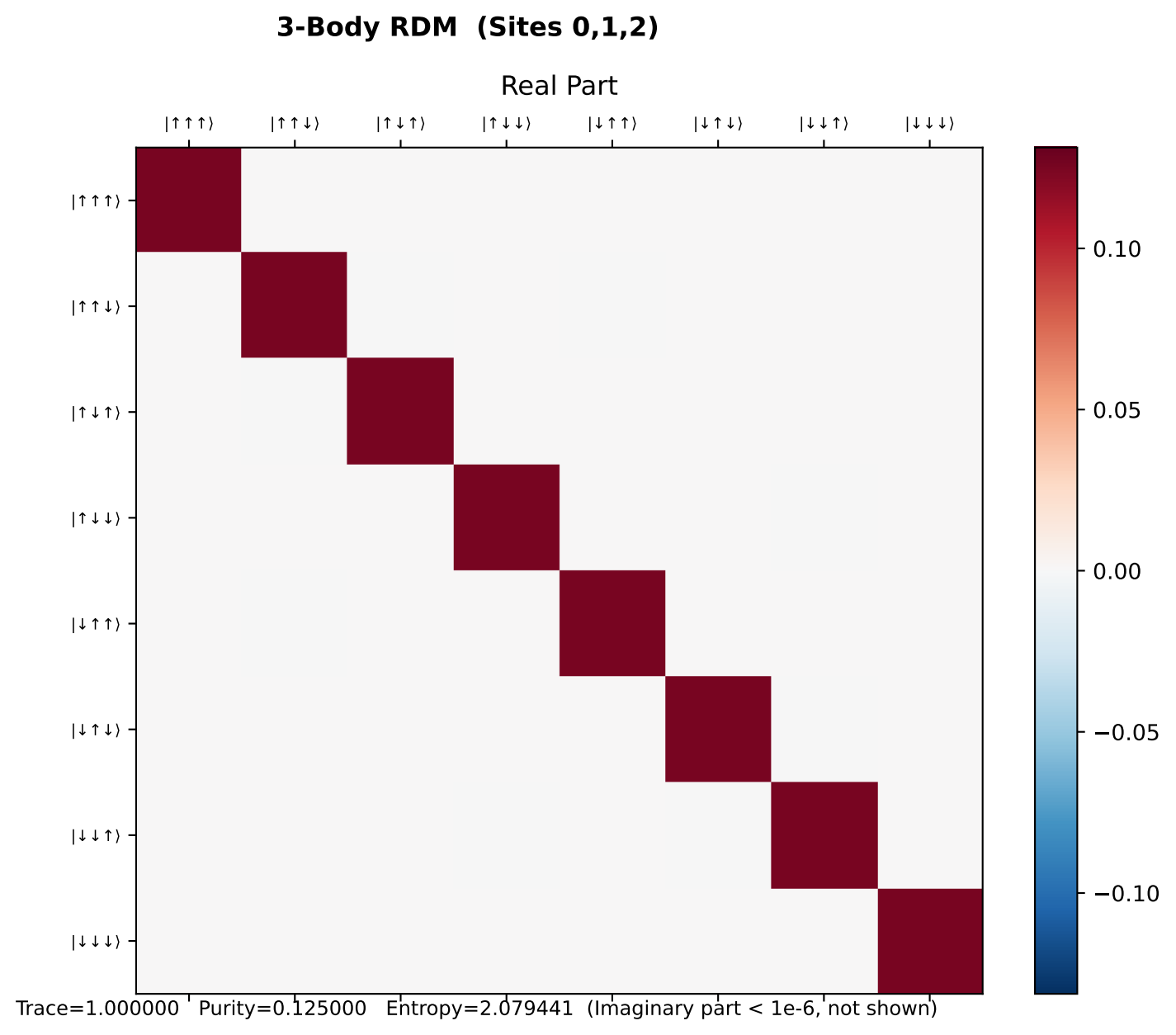
THIS PDF

Body order : 3
Clusters   : 20

Date: 2026-07-10 17:23

Summary — 3-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2	0.125000	—	0.000000	0.007884	—	0.422170	STRONGLY MIXED
0,1,3	0.148894	—	0.000000	0.000024	—	0.315508	STRONGLY MIXED
0,1,4	0.125559	—	0.000000	0.011454	—	0.405751	STRONGLY MIXED
0,1,5	0.125067	—	0.000000	0.013866	—	0.417046	STRONGLY MIXED
0,2,3	0.125558	—	0.000000	0.001181	—	0.405758	STRONGLY MIXED
0,2,4	0.148907	—	0.000000	0.000047	—	0.310329	STRONGLY MIXED
0,2,5	0.125065	—	0.000000	0.004814	—	0.417145	STRONGLY MIXED
0,3,4	0.125949	—	0.000000	0.005155	—	0.443271	STRONGLY MIXED
0,3,5	0.150430	—	0.000000	0.000207	—	0.315200	STRONGLY MIXED
0,4,5	0.150975	—	0.000000	0.000649	—	0.314028	STRONGLY MIXED
1,2,3	0.496220	—	0.000000	0.029778	—	0.089519	MIXED
1,2,4	0.486923	—	0.000000	0.001094	—	0.093802	MIXED
1,2,5	0.125000	—	0.000000	0.000099	—	0.422828	STRONGLY MIXED
1,3,4	0.499664	—	0.000000	0.070670	—	0.087949	MIXED
1,3,5	0.145172	—	0.000000	0.000241	—	0.350783	STRONGLY MIXED
1,4,5	0.125941	—	0.000000	0.000005	—	0.445233	STRONGLY MIXED
2,3,4	0.499777	—	0.000000	0.073893	—	0.087898	MIXED
2,3,5	0.125967	—	0.000000	0.007822	—	0.445647	STRONGLY MIXED
2,4,5	0.144255	—	0.000000	0.007474	—	0.353026	STRONGLY MIXED
3,4,5	0.125368	—	0.000000	0.000229	—	0.413806	STRONGLY MIXED


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250000348
  Von Neumann entropy  : 2.0794414025
  Eigenvalue min       : 1.248858e-01
  Eigenvalue max       : 1.250381e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12500003

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

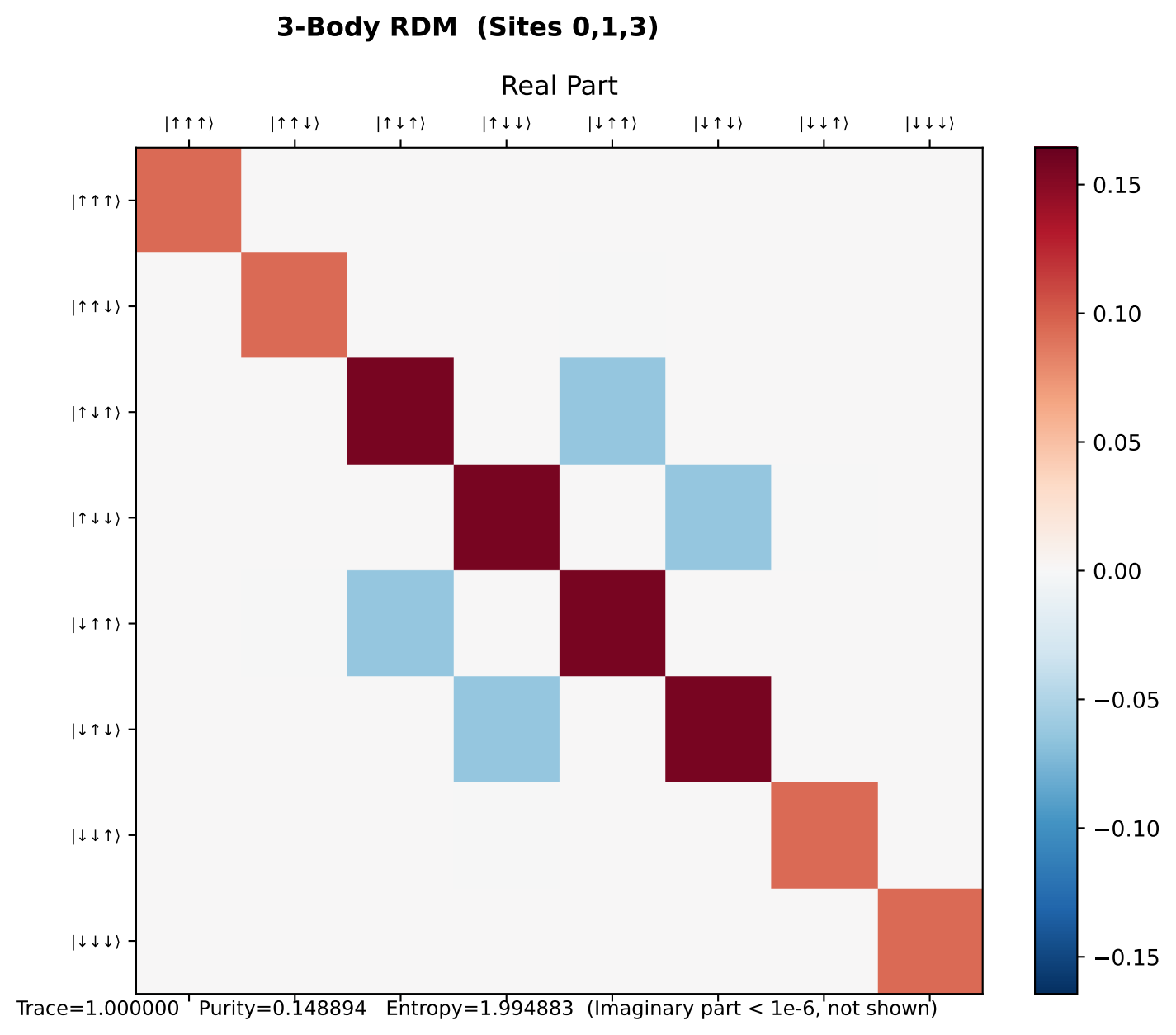
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.42216972

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.092203  C_GME=0.065668
2. p=0.103325  C_GME=0.002702
3. p=0.100848  C_GME=0.000000
4. p=0.106233  C_GME=0.000000
5. p=0.089149  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,1,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148894 Entropy=1.994883 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1488944497
  Von Neumann entropy  : 1.9948828658
  Eigenvalue min       : 9.344683e-02
  Eigenvalue max       : 2.196595e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14889445

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

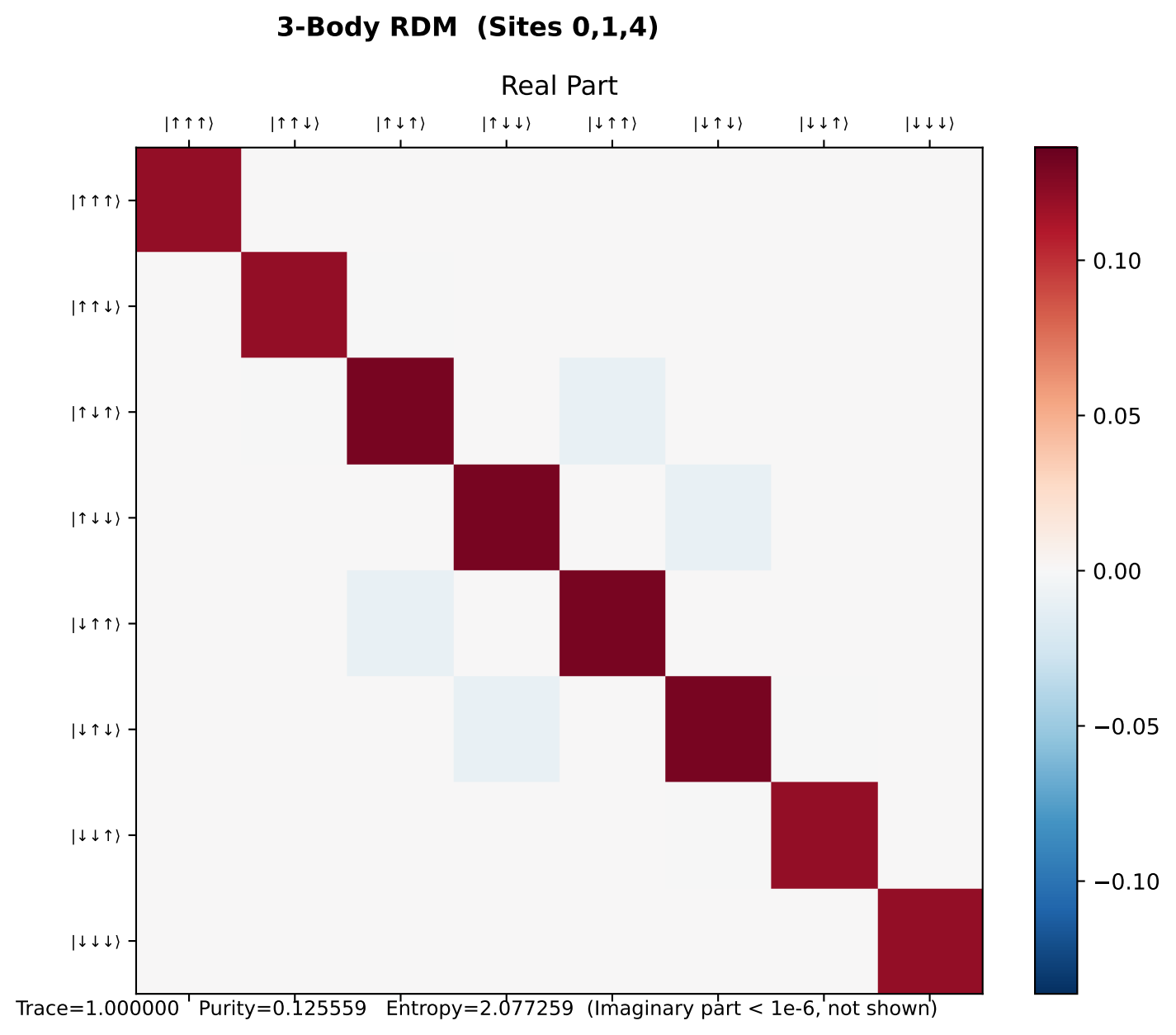
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.31550813

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.120059  C_GME=0.000000
2. p=0.088648  C_GME=0.000079
3. p=0.104480  C_GME=0.000001
4. p=0.121740  C_GME=0.000001
5. p=0.072438  C_GME=0.000001
... and 5 more states
```


3-Body RDM (Sites 0,1,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125559 Entropy=2.077259 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1255590621
Von Neumann entropy	2.0772592054
Eigenvalue min	1.201735e-01
Eigenvalue max	1.394792e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.12555906

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

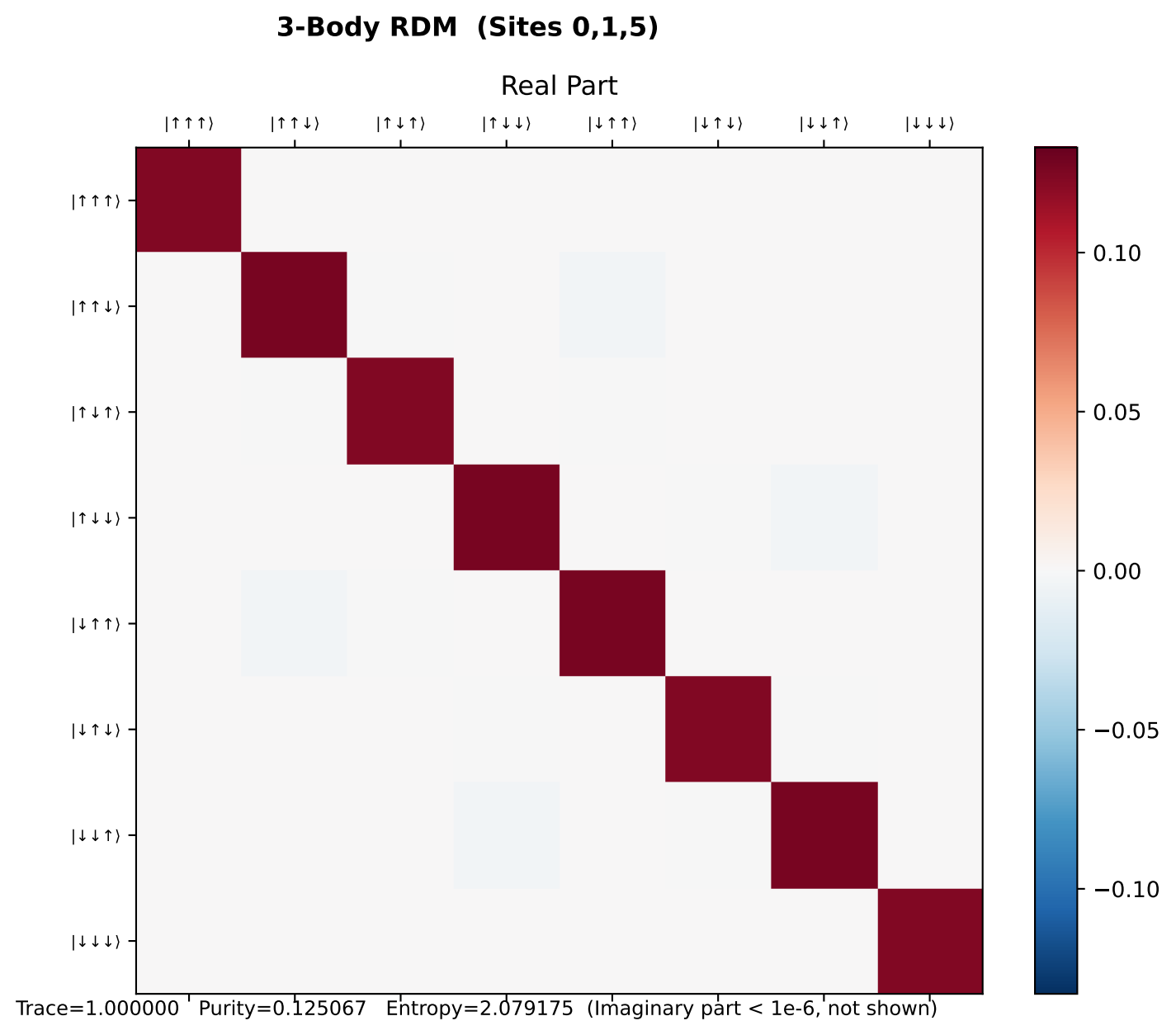
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.40575139

Status: inconclusive

Solver source:
–

1. p=0.095591 C_GME=0.000000
2. p=0.088242 C_GME=0.000001
3. p=0.086305 C_GME=0.000000
4. p=0.091520 C_GME=0.000000
5. p=0.103396 C_GME=0.000000
... and 5 more states


3-Body RDM (Sites 0,1,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125067 Entropy=2.079175 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250673282
  Von Neumann entropy  : 2.0791745511
  Eigenvalue min       : 1.231782e-01
  Eigenvalue max       : 1.300146e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12506733

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

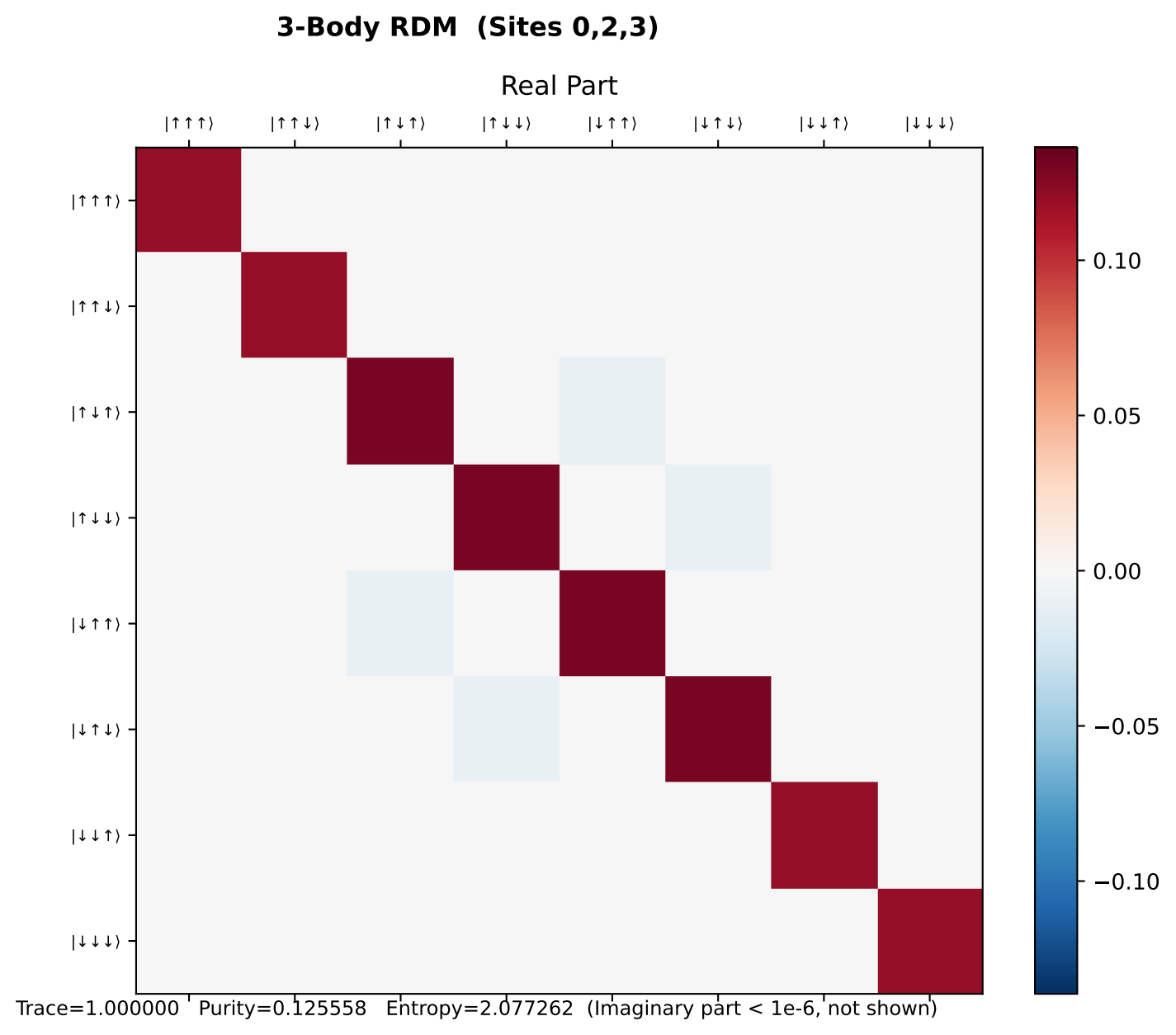
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.41704637

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.109569  C_GME=0.012480
2. p=0.114659  C_GME=0.029557
3. p=0.100762  C_GME=0.000001
4. p=0.102919  C_GME=0.000000
5. p=0.106134  C_GME=0.002092
... and 5 more states
```


3-Body RDM (Sites 0,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125558 Entropy=2.077262 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1255583975
  Von Neumann entropy  : 2.0772617699
  Eigenvalue min       : 1.201763e-01
  Eigenvalue max       : 1.394706e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12555840

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

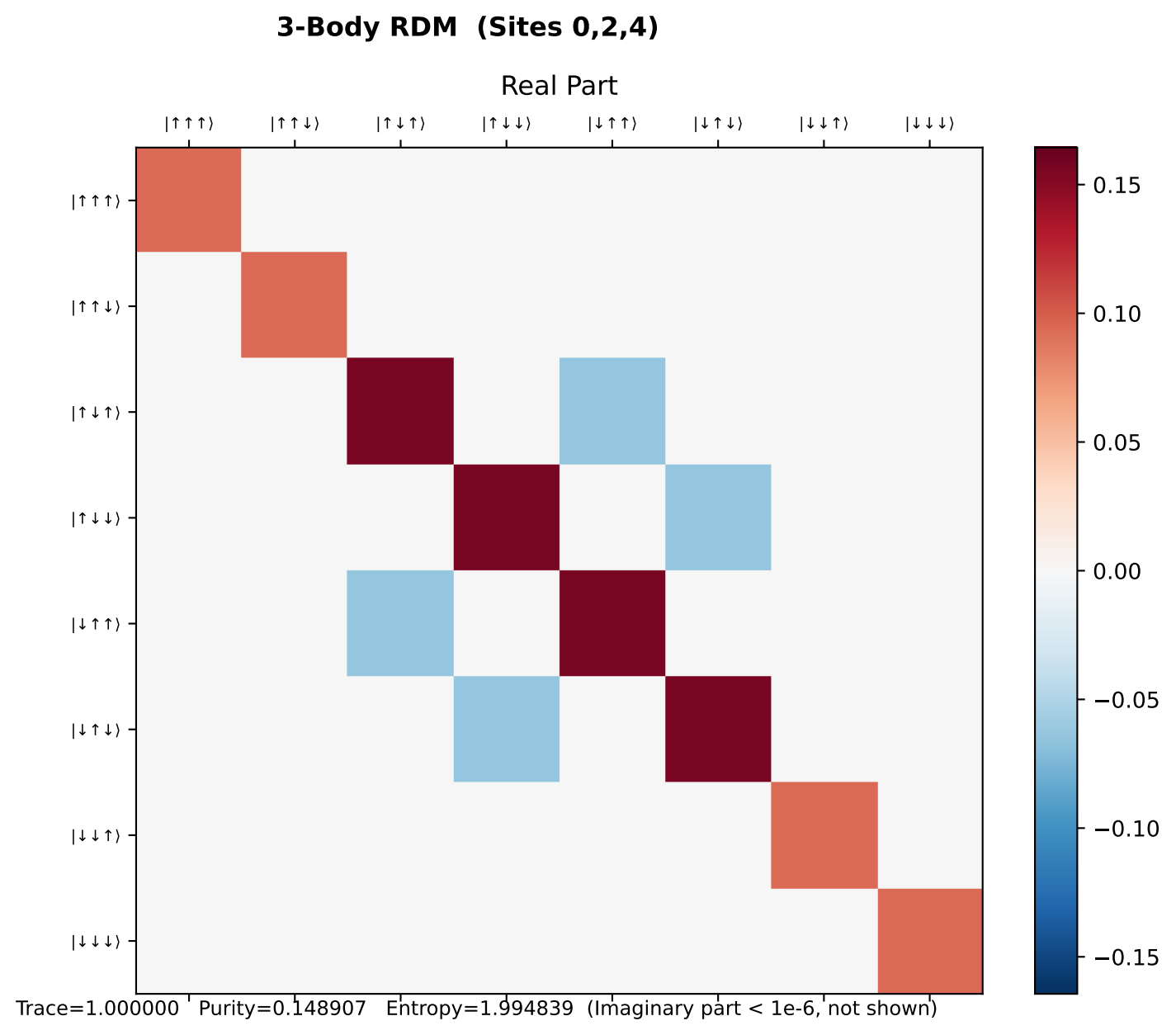
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.40575797

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.081223  C_GME=0.000000
2. p=0.068643  C_GME=0.000000
3. p=0.106130  C_GME=0.000000
4. p=0.106849  C_GME=0.000201
5. p=0.119660  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.148907 Entropy=1.994839 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1489072792
  Von Neumann entropy  : 1.9948394271
  Eigenvalue min       : 9.343835e-02
  Eigenvalue max       : 2.196849e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.14890728

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

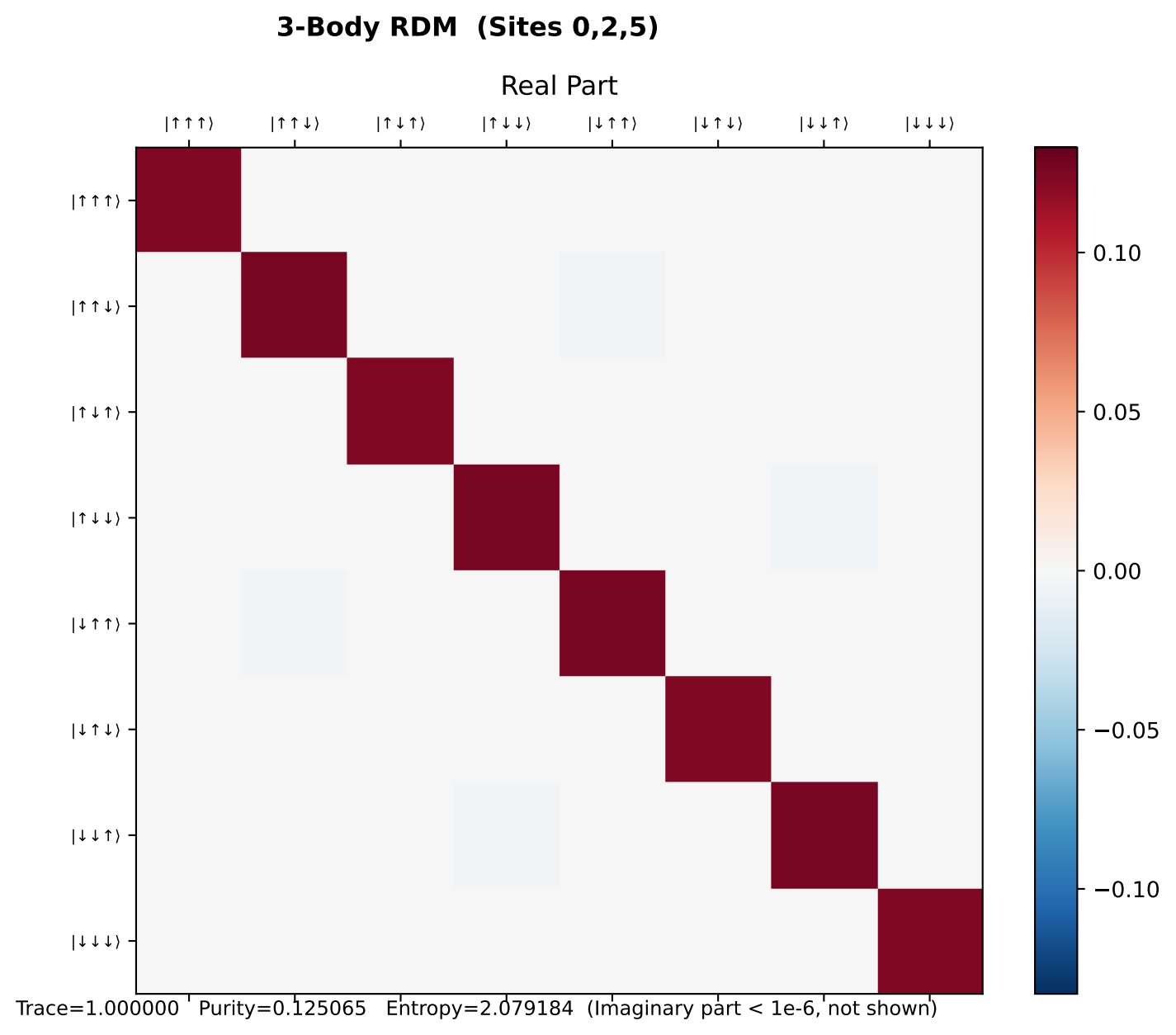
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.31032906

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.099262  C_GME=0.000000
2. p=0.083932  C_GME=0.000000
3. p=0.072399  C_GME=0.000002
4. p=0.097861  C_GME=0.000000
5. p=0.106733  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,2,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125065 Entropy=2.079184 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,2,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250649358
  Von Neumann entropy  : 2.0791839988
  Eigenvalue min       : 1.232101e-01
  Eigenvalue max       : 1.299246e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12506494

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

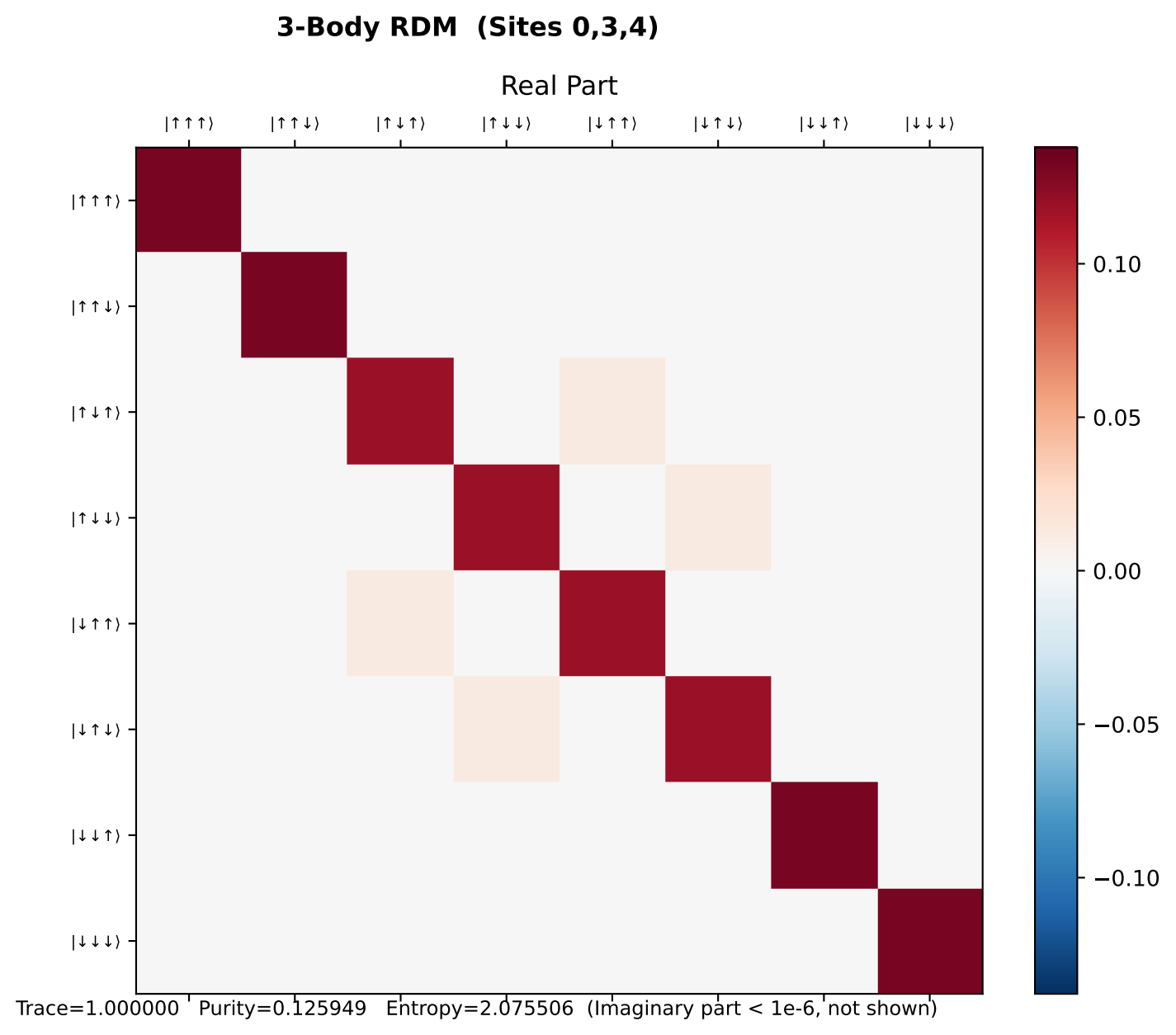
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.41714457

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.109169  C_GME=0.002594
2. p=0.113396  C_GME=0.003989
3. p=0.096414  C_GME=0.000006
4. p=0.072315  C_GME=0.000000
5. p=0.120639  C_GME=0.006020
... and 5 more states
```


3-Body RDM (Sites 0,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125949 Entropy=2.075506 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259488965
  Von Neumann entropy  : 2.0755063875
  Eigenvalue min       : 1.061364e-01
  Eigenvalue max       : 1.312880e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12594890

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
------------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

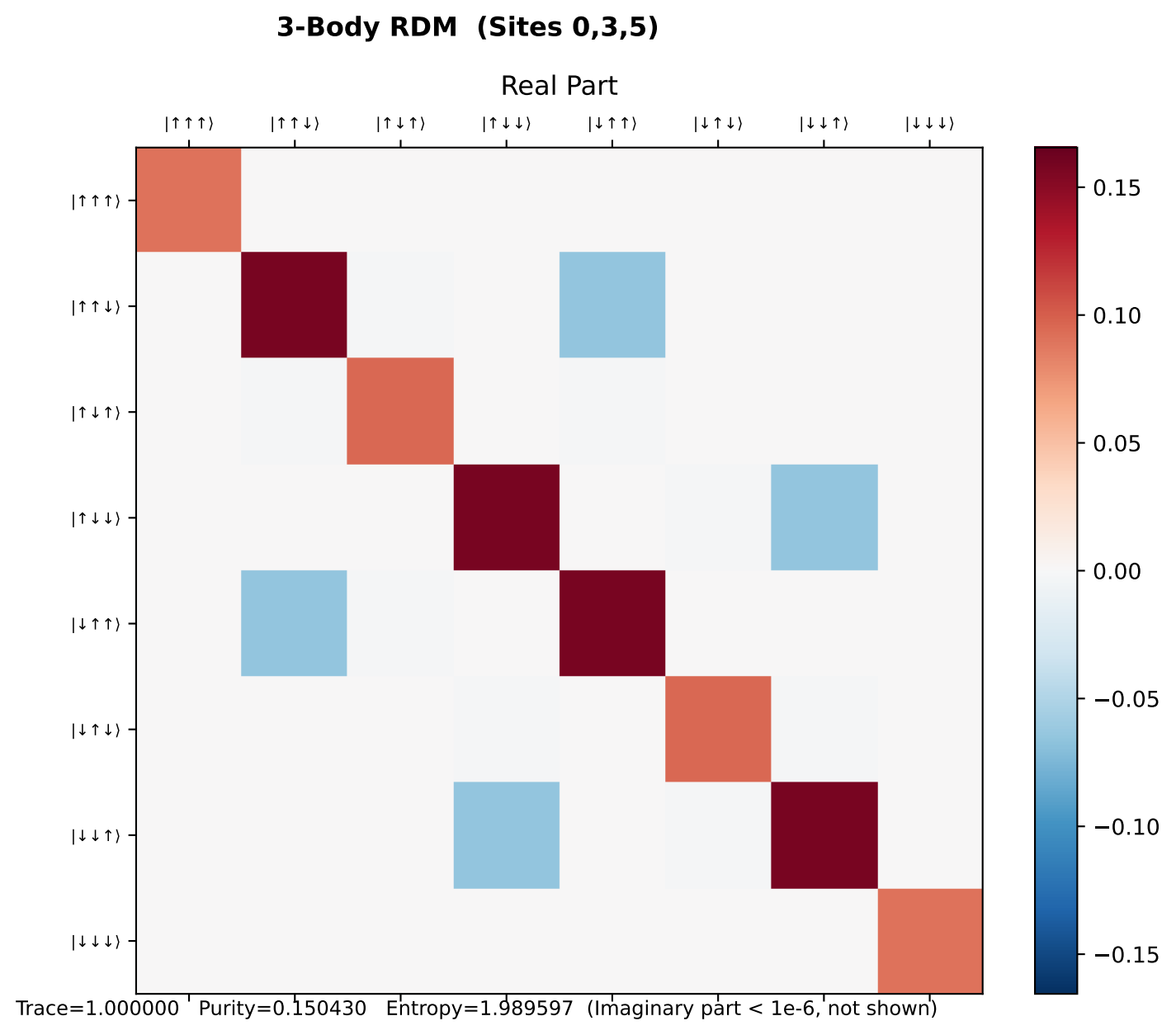
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.44327108

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.121226  C_GME=0.000000
2. p=0.083536  C_GME=0.033624
3. p=0.101641  C_GME=0.000011
4. p=0.099296  C_GME=0.002250
5. p=0.118199  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 0,3,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.150430 Entropy=1.989597 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,3,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1504300892
  Von Neumann entropy  : 1.9895972634
  Eigenvalue min       : 9.049354e-02
  Eigenvalue max       : 2.225630e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.15043009

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

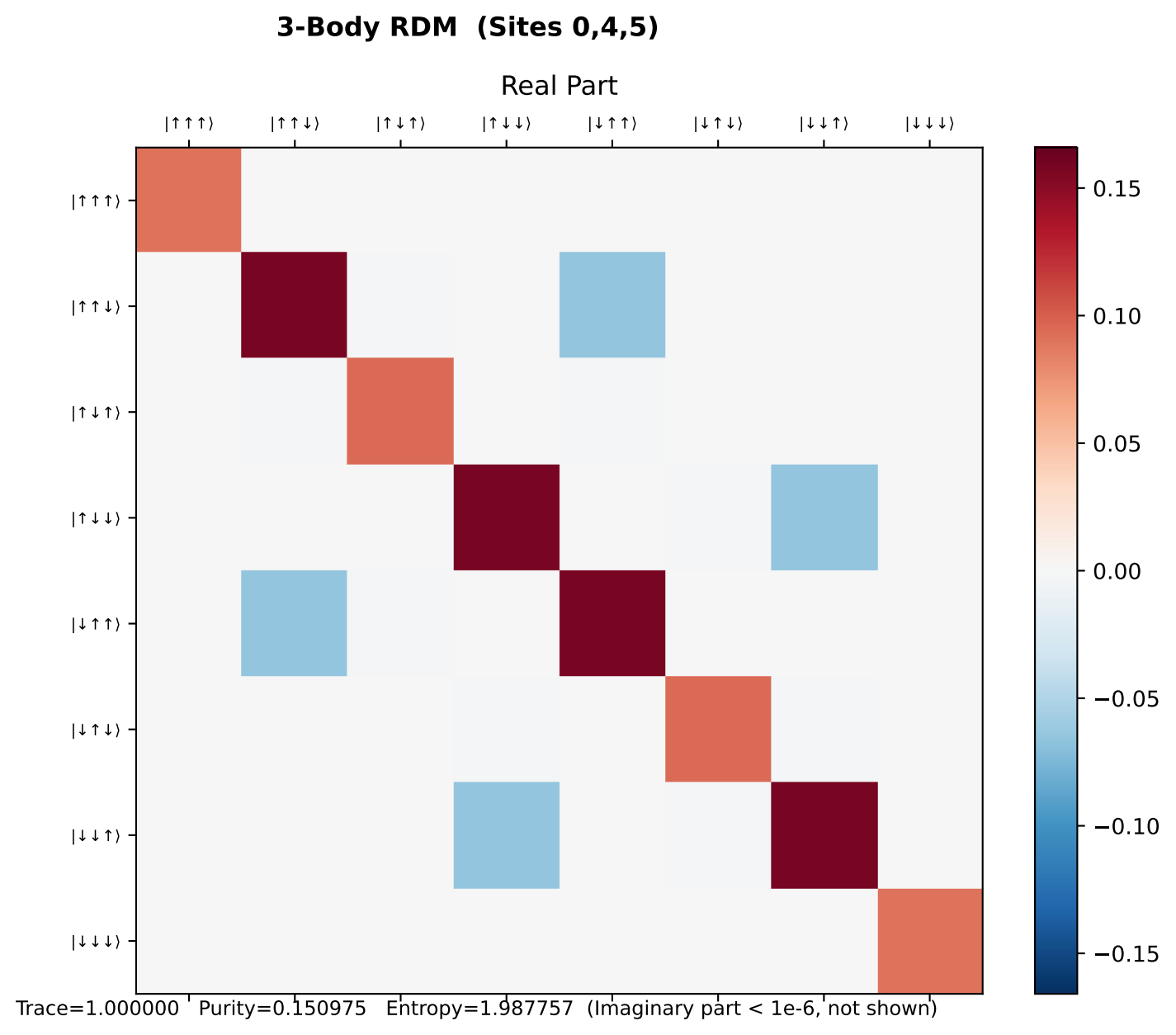
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.31520007

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.103582  C_GME=0.000000
2. p=0.089920  C_GME=0.000000
3. p=0.092729  C_GME=0.001958
4. p=0.121128  C_GME=0.000000
5. p=0.111459  C_GME=0.000225
... and 5 more states
```


3-Body RDM (Sites 0,4,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.150975 Entropy=1.987757 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1509750785
Von Neumann entropy	1.9877574325
Eigenvalue min	9.014436e-02
Eigenvalue max	2.236047e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.15097508

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

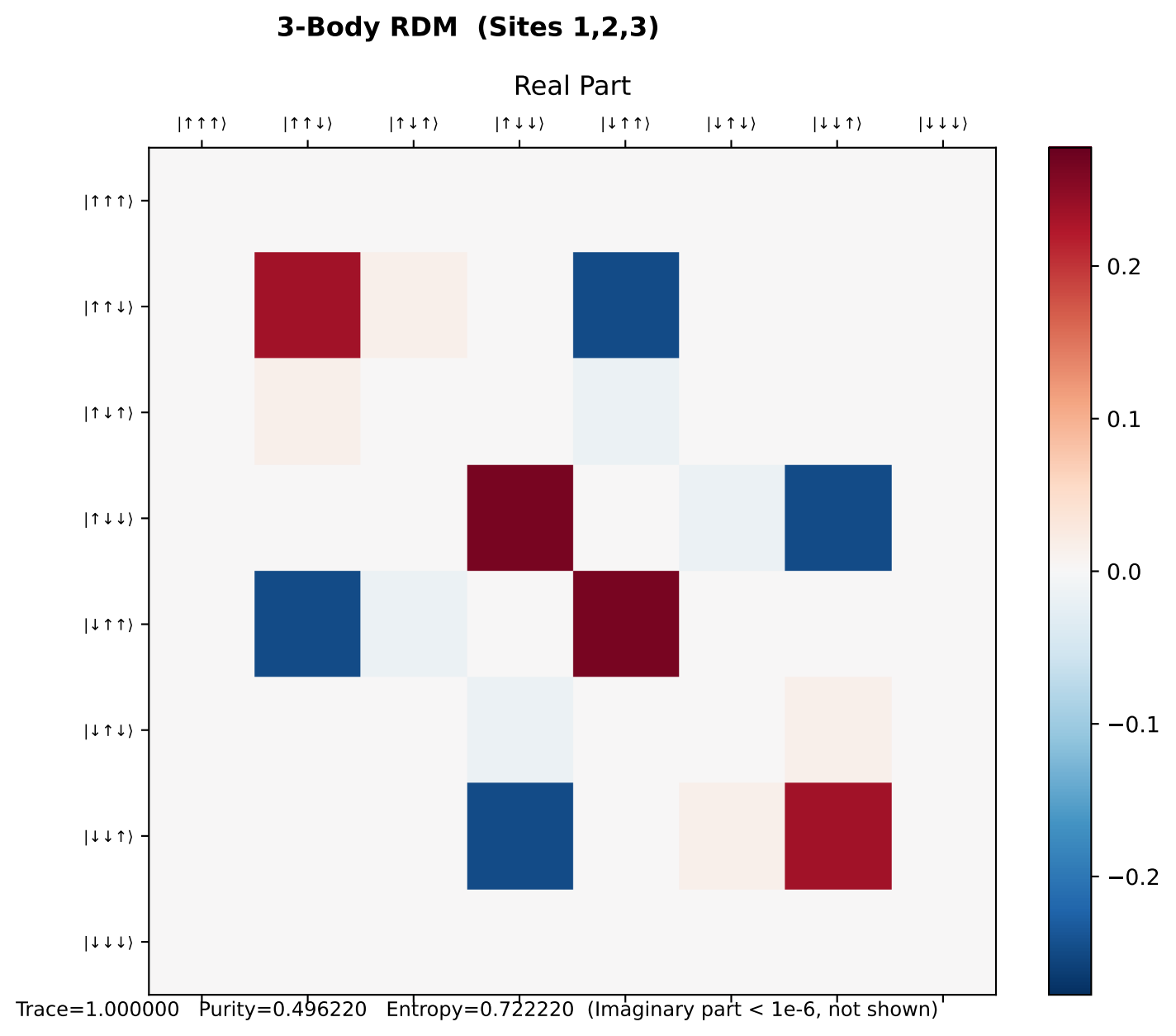
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.31402822

Status: inconclusive

Solver source:
–

1. p=0.095183 C_GME=0.000000
2. p=0.070629 C_GME=0.000000
3. p=0.098901 C_GME=0.000000
4. p=0.116152 C_GME=0.001117
5. p=0.097249 C_GME=0.000000
... and 5 more states


3-Body RDM (Sites 1,2,3)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.496220 Entropy=0.722220 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,3)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.4962202856
Von Neumann entropy	0.7222196733
Eigenvalue min	6.299232e-04
Eigenvalue max	4.981054e-01
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.49622029

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.

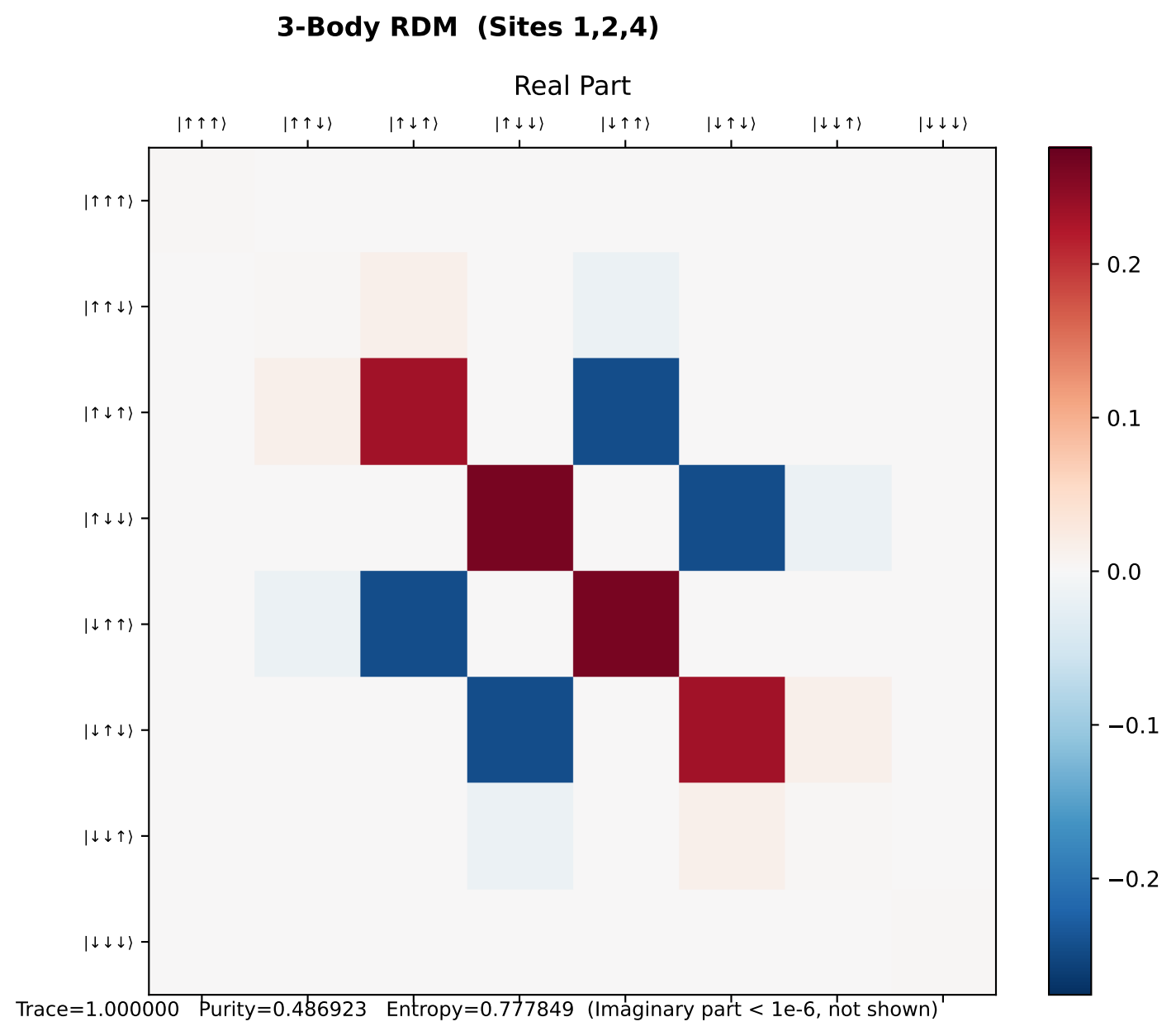
Convex Roof Analysis
==========

Lower Bounds:
C_LB = 0.00000000
C_GME (unitary optimization):
C_stiefel (unitary opt) = nan BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
C_UB (eigendecomp) = 0.08951890

Status: inconclusive

Solver source:
–

1. p=0.157725 C_GME=0.007716
2. p=0.013323 C_GME=0.314972
3. p=0.040591 C_GME=0.162608
4. p=0.159436 C_GME=0.009797
5. p=0.036064 C_GME=0.125376
... and 5 more states


3-Body RDM (Sites 1,2,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.486923 Entropy=0.777849 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,4)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4869226154
  Von Neumann entropy  : 0.7778492876
  Eigenvalue min       : 2.197277e-03
  Eigenvalue max       : 4.934033e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.48692262

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

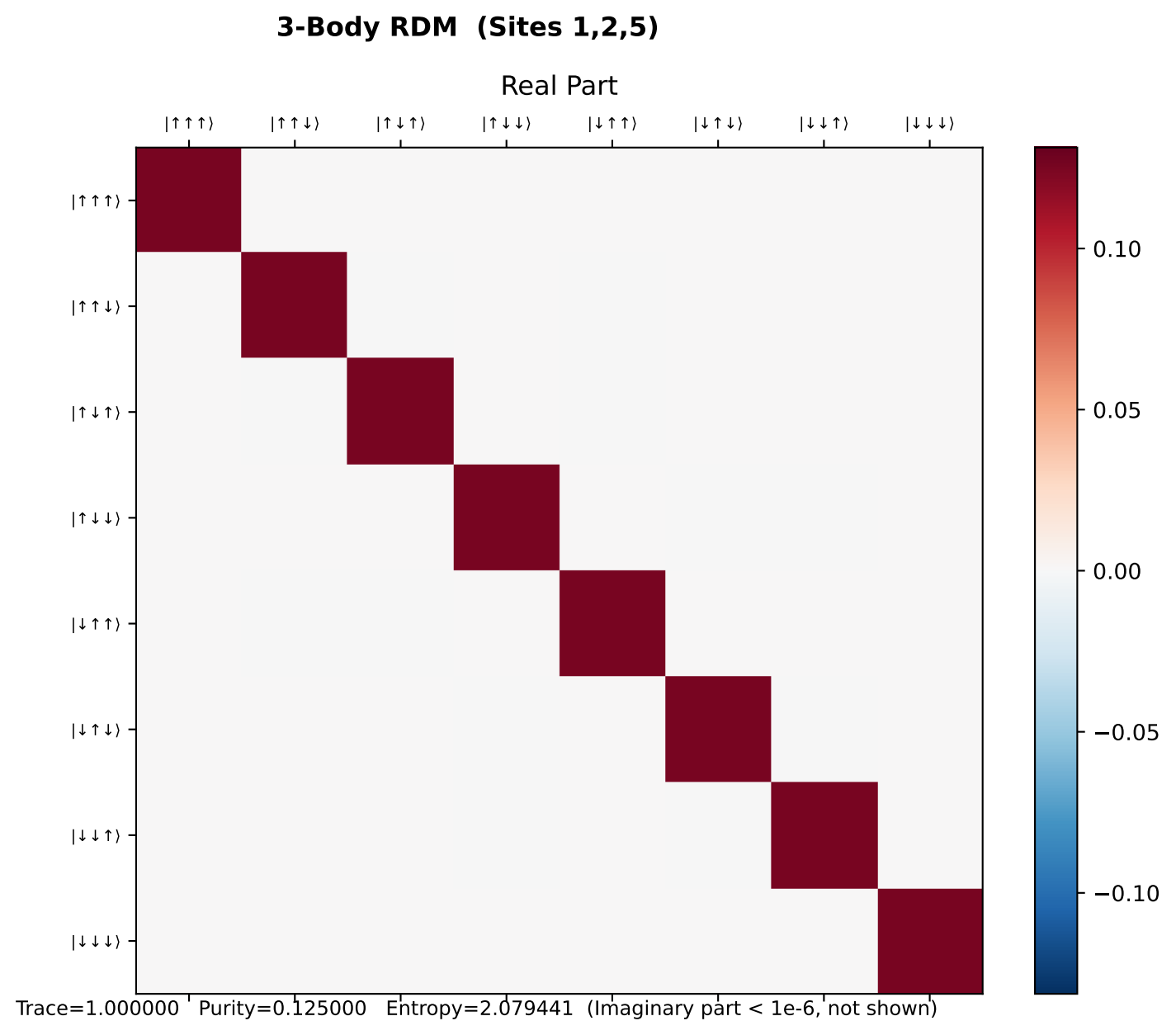
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.09380175

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.164470  C_GME=0.000002
2. p=0.030433  C_GME=0.032383
3. p=0.092984  C_GME=0.000003
4. p=0.002223  C_GME=0.019633
5. p=0.204061  C_GME=0.000002
... and 5 more states
```


3-Body RDM (Sites 1,2,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1250001535
  Von Neumann entropy  : 2.0794409280
  Eigenvalue min       : 1.249047e-01
  Eigenvalue max       : 1.252374e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.12500015

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

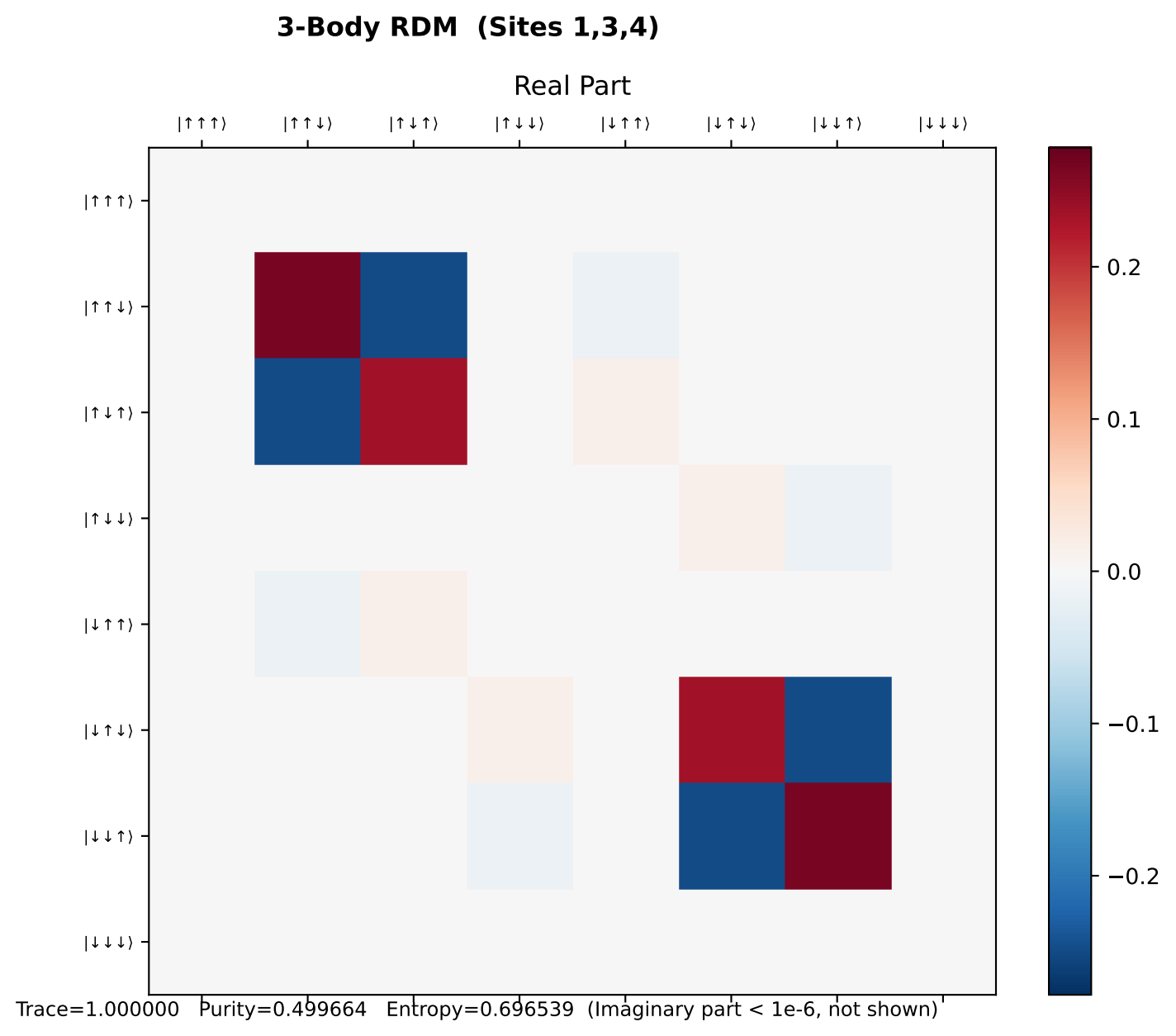
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.42282837

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.105240  C_GME=0.000000
2. p=0.107961  C_GME=0.000512
3. p=0.083704  C_GME=0.000000
4. p=0.100358  C_GME=0.000413
5. p=0.090935  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 1,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499664 Entropy=0.696539 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4996641682
  Von Neumann entropy  : 0.6965387921
  Eigenvalue min       : 5.510387e-05
  Eigenvalue max       : 4.998320e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.49966417

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

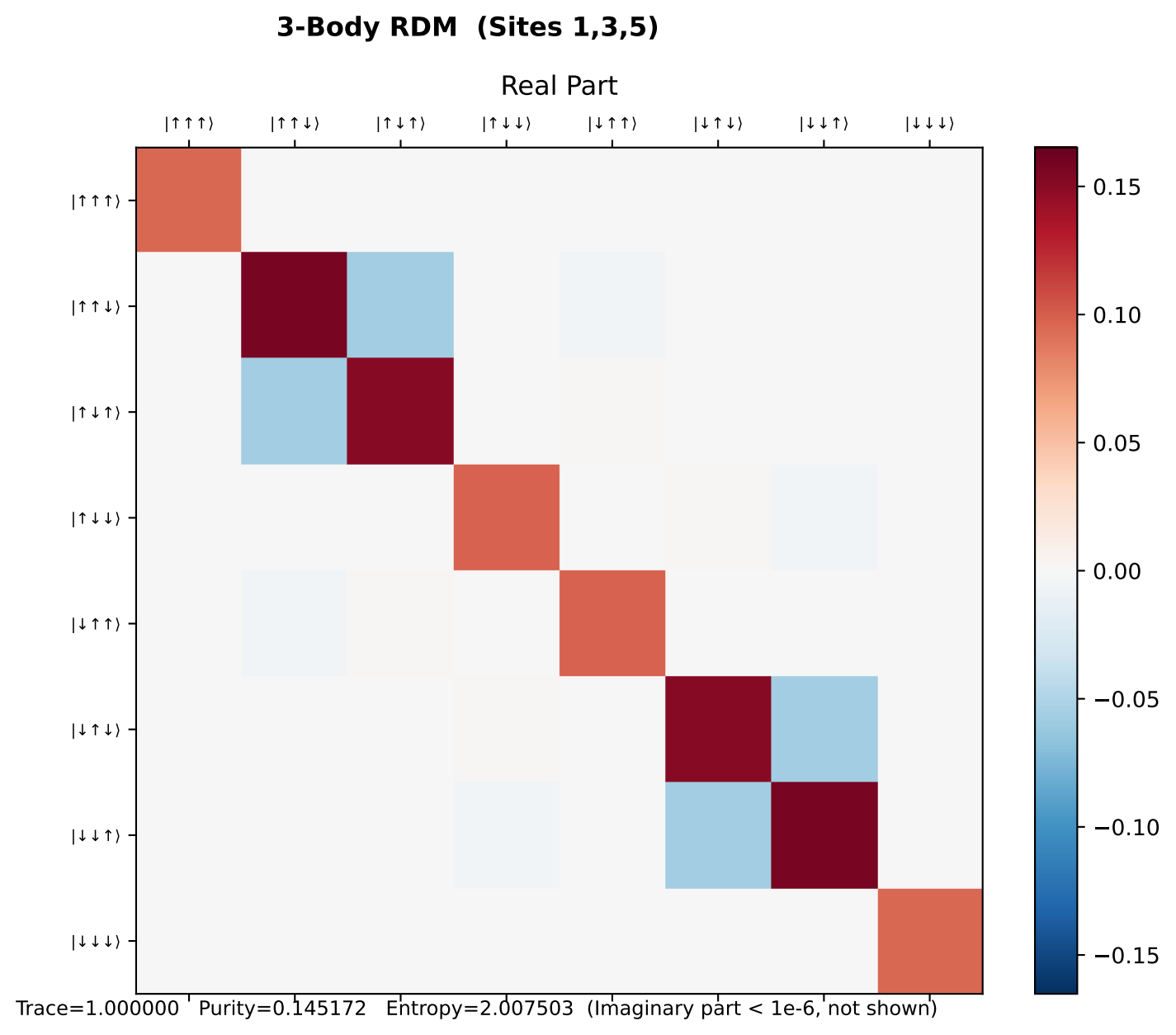
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08794920

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.123727  C_GME=0.053650
2. p=0.165834  C_GME=0.072897
3. p=0.054459  C_GME=0.113968
4. p=0.165610  C_GME=0.068434
5. p=0.155206  C_GME=0.078476
... and 4 more states
```


3-Body RDM (Sites 1,3,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.145172 Entropy=2.007503 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,3,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1451716469
  Von Neumann entropy  : 2.0075033507
  Eigenvalue min       : 9.470402e-02
  Eigenvalue max       : 2.119283e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14517165

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

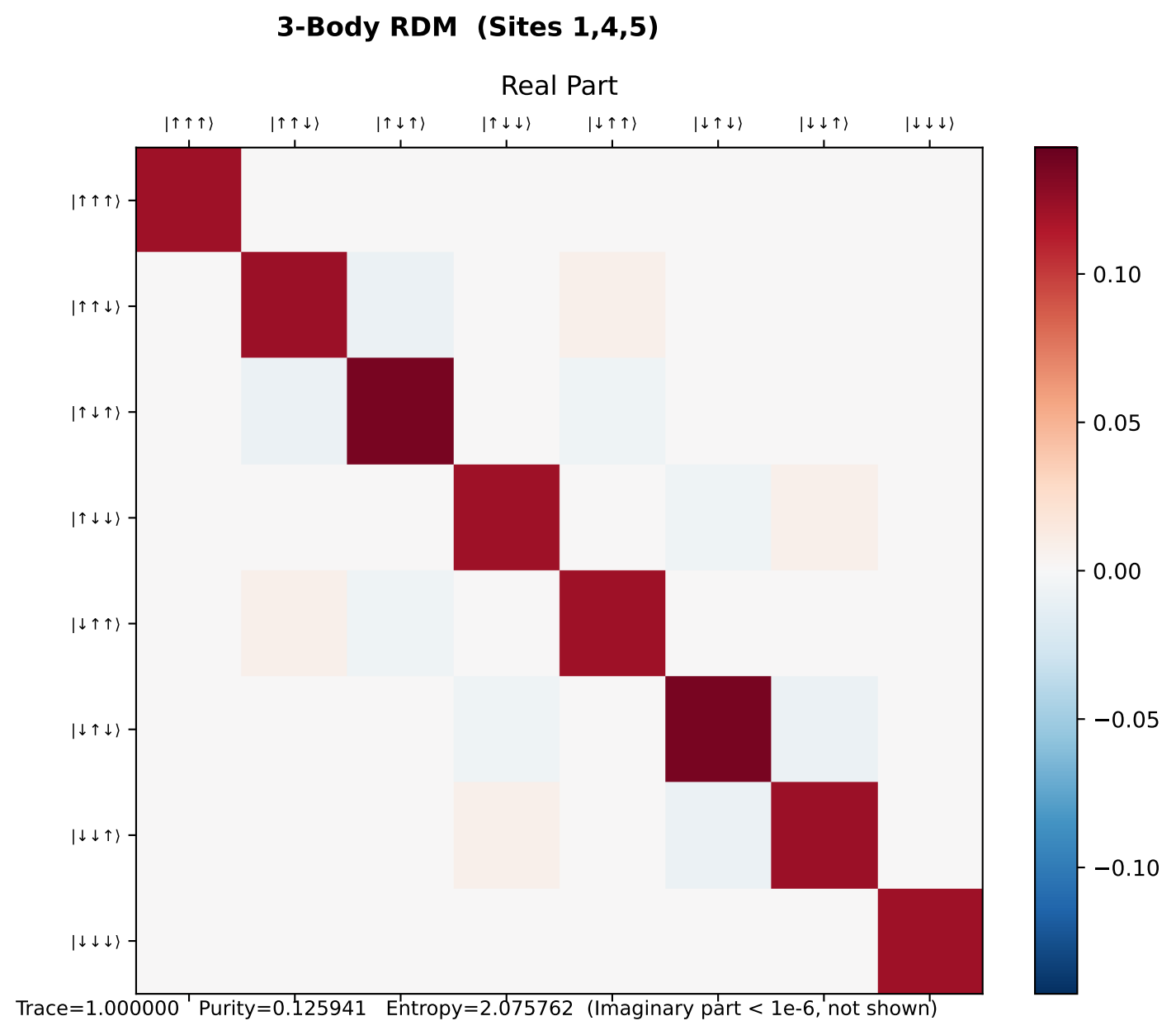
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.35078290

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.094596  C_GME=0.000000
2. p=0.086184  C_GME=0.000000
3. p=0.096792  C_GME=0.000000
4. p=0.060065  C_GME=0.000000
5. p=0.086946  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 1,4,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125941 Entropy=2.075762 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,4,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259414218
  Von Neumann entropy  : 2.0757624647
  Eigenvalue min       : 1.141110e-01
  Eigenvalue max       : 1.430661e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.12594142

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

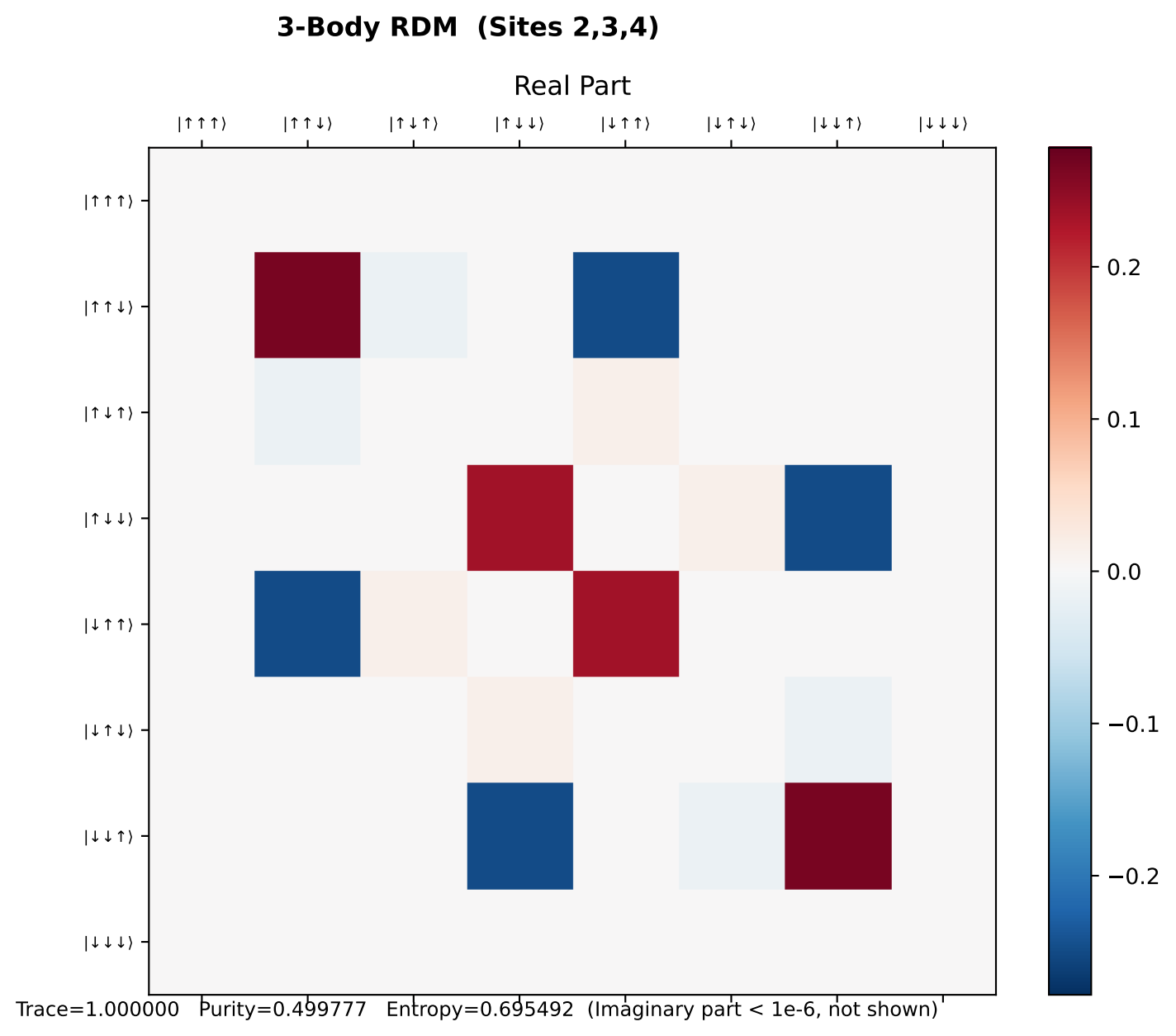
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.44523254

  Status: no_GME

  Solver source:
    –
```

```
1. p=0.111628  C_GME=0.000000
2. p=0.075521  C_GME=0.000000
3. p=0.107830  C_GME=0.000000
4. p=0.115988  C_GME=0.000004
5. p=0.076211  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 2,3,4)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.499777 Entropy=0.695492 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,3,4)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.4997768051
  Von Neumann entropy  : 0.6954922543
  Eigenvalue min       : 3.633686e-05
  Eigenvalue max       : 4.998884e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.49977681

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

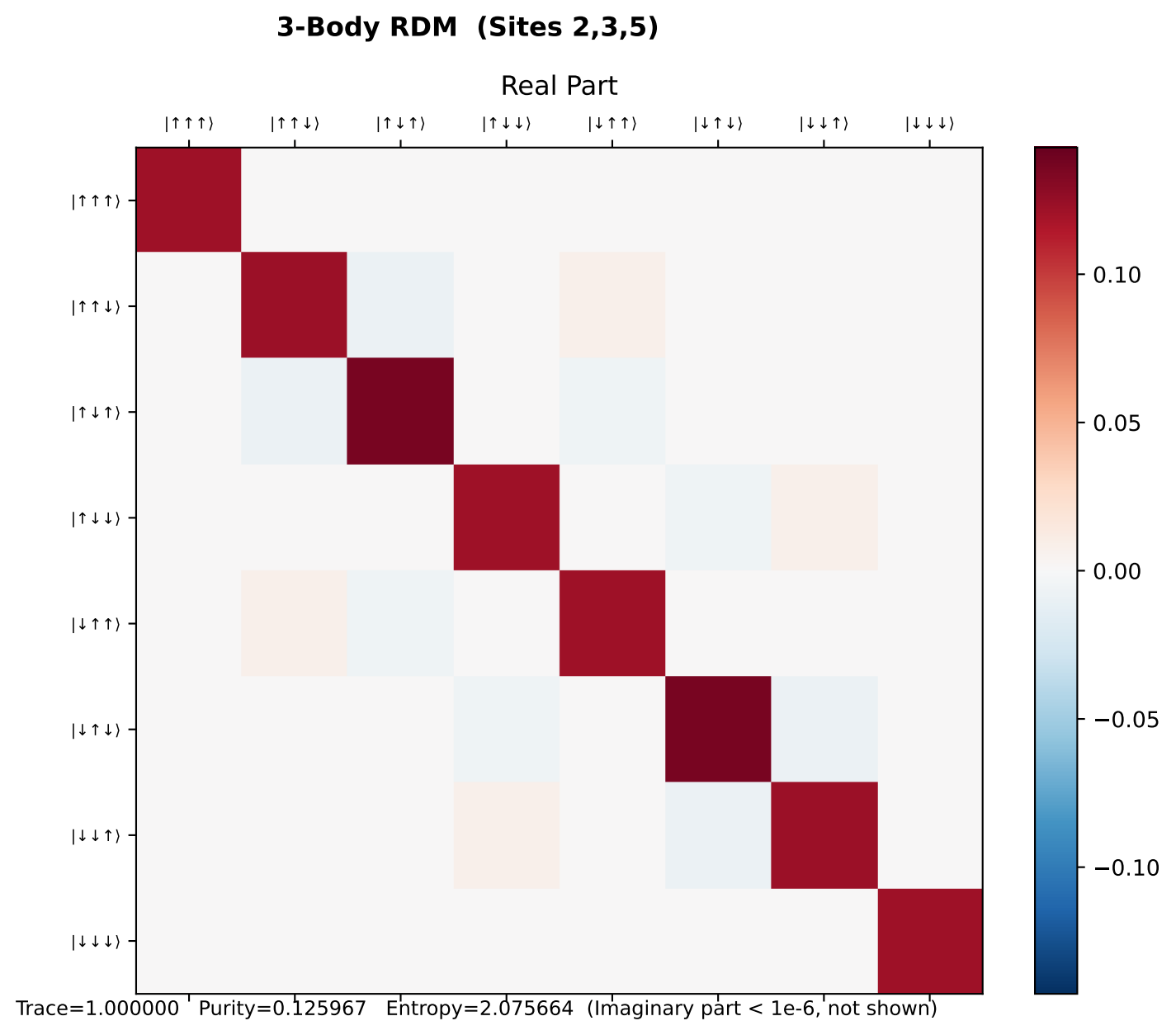
```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.08789793

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.136419  C_GME=0.082444
2. p=0.098914  C_GME=0.094233
3. p=0.144729  C_GME=0.072565
4. p=0.087844  C_GME=0.057355
5. p=0.036166  C_GME=0.040610
... and 4 more states
```


3-Body RDM (Sites 2,3,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125967 Entropy=2.075664 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,3,5)

```
RDM Metrics
===================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1259673001
  Von Neumann entropy  : 2.0756636393
  Eigenvalue min       : 1.140523e-01
  Eigenvalue max       : 1.433427e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
===================================

  gamma = Tr(rho^2) = 0.12596730

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-----------------------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

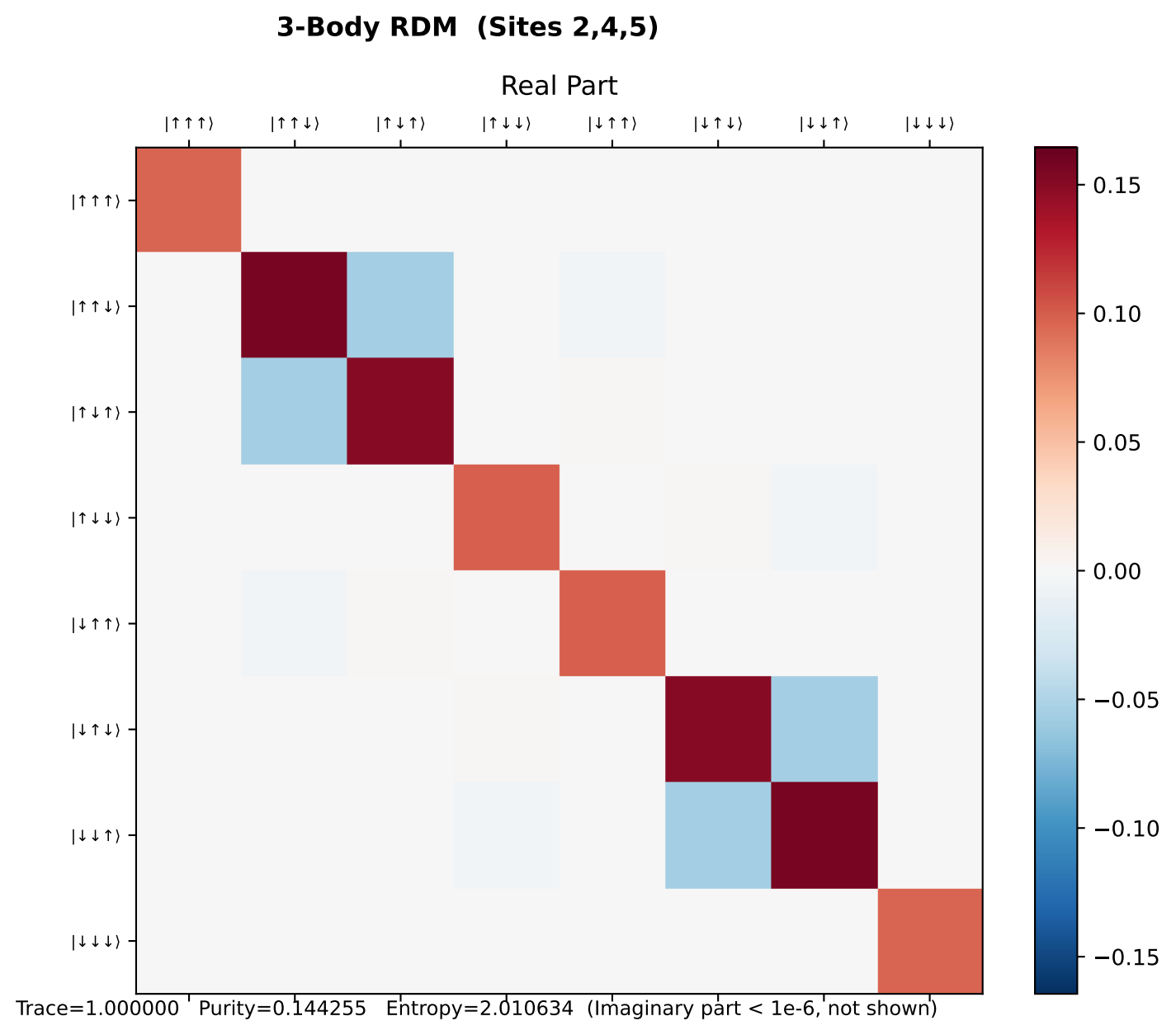
```
Convex Roof Analysis
===================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.44564732

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.091854  C_GME=0.000000
2. p=0.097901  C_GME=0.000000
3. p=0.093619  C_GME=0.083079
4. p=0.105023  C_GME=0.000000
5. p=0.117633  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 2,4,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.144255 Entropy=2.010634 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1442552519
  Von Neumann entropy  : 2.0106344236
  Eigenvalue min       : 9.531950e-02
  Eigenvalue max       : 2.099249e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.14425525

  A            | B            | Tr(rho_A^2)
  ____________ | ____________ | ___________

STATUS: MIXED STATE
____________________________________

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

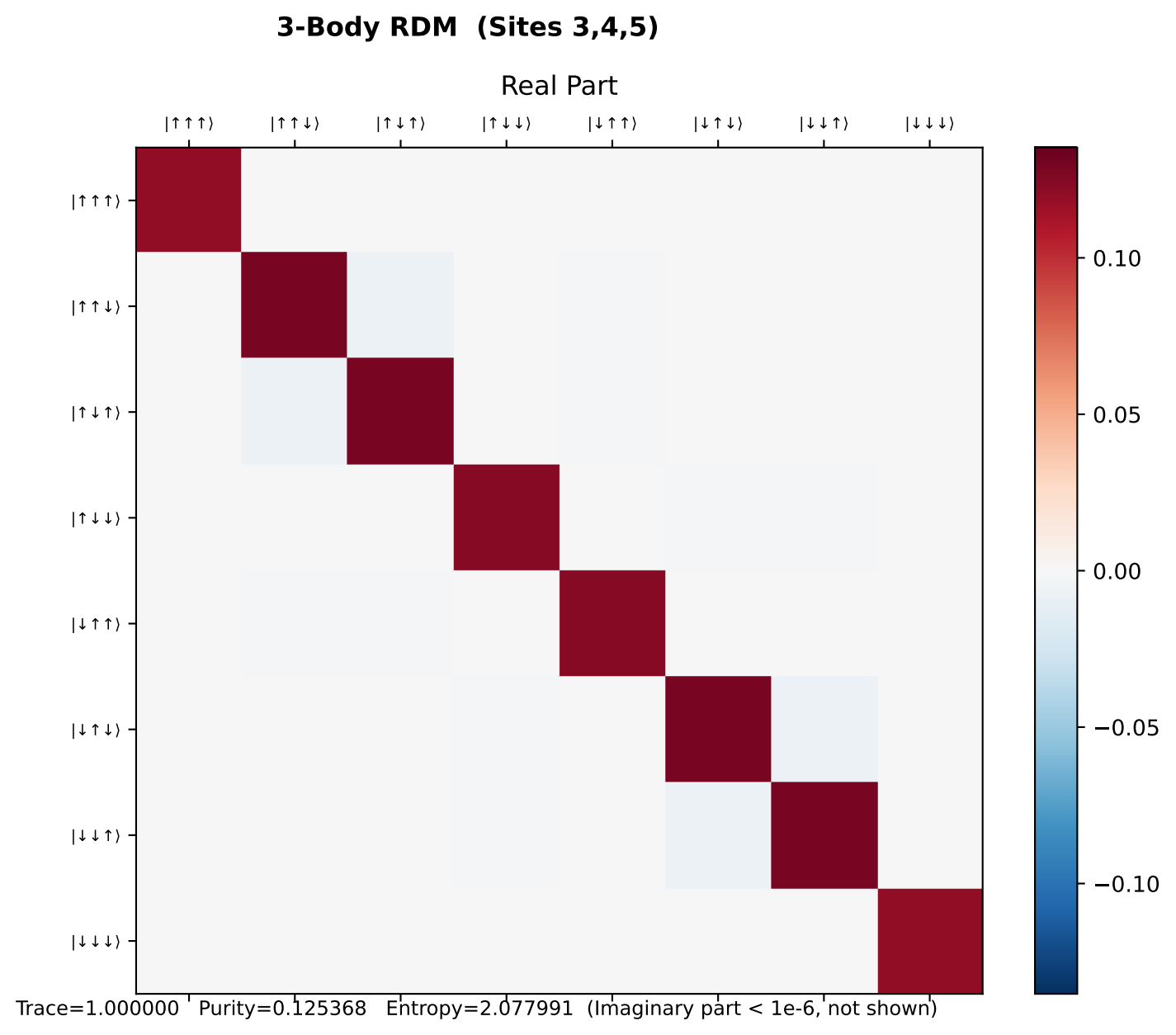
```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.35302550

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.144713  C_GME=0.000274
2. p=0.098395  C_GME=0.000000
3. p=0.098775  C_GME=0.001334
4. p=0.087790  C_GME=0.000000
5. p=0.098204  C_GME=0.000000
... and 5 more states
```


3-Body RDM (Sites 3,4,5)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.125368 Entropy=2.077991 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 3,4,5)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 0.1253680220
  Von Neumann entropy  : 2.0779912025
  Eigenvalue min       : 1.195895e-01
  Eigenvalue max       : 1.361948e-01
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 0.12536802

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------

STATUS: MIXED STATE
-------------------

Tr(rho^2) < 1. Pure-state C_GME
not applicable. See convex-roof
result below.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = nan   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.41380586

  Status: inconclusive

  Solver source:
    –
```

```
1. p=0.108250  C_GME=0.000001
2. p=0.111908  C_GME=0.000575
3. p=0.088163  C_GME=0.000942
4. p=0.116013  C_GME=0.000000
5. p=0.089636  C_GME=0.000001
... and 5 more states
```